1 Additional Baselines

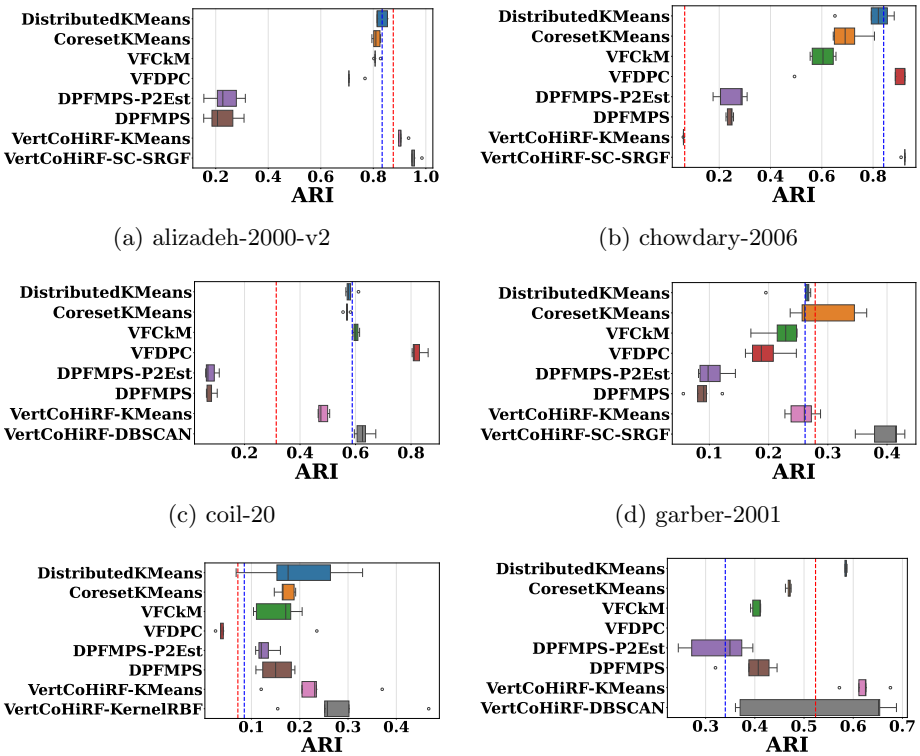
1.1 Additional Baselines on Multimodal Dataset

Table 1: ARI score on the synthetic multimodal dataset.

Method	Agent_0	Agent_1	Mean
K-Means	0.075216	0.275385	0.175301
CoHiRF-KMeans	0.042929	0.571222	0.307075
Distr.KMeans	<u>0.430624</u>	0.430624	<u>0.430624</u>
CoresetKMeans	0.000911	0.468107	0.234509
DPFMPS	0.114054	0.555552	0.334803
DPFMPS-2PEst	0.056390	<u>0.565406</u>	0.310898
VFCkM	0.001302	0.441553	0.221427
VFDPC	**1**	**1**	**1**
VertCoHiRF	**1**	**1**	**1**

1.2 Additional Baselines on Real-World Datasets

1.2.1 ARI



(a) alizadeh-2000-v2

(b) chowdary-2006

(c) coil-20

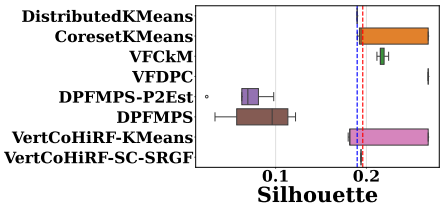
(d) garber-2001

(e) nursery

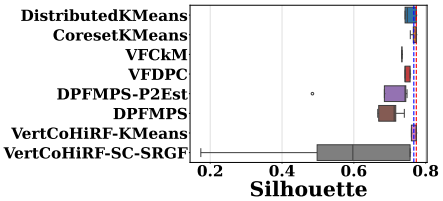
(f) shuttle

Figure 1: ARI boxplots across feature partitions. The blue and red dashed line shows the non-collaborative local k-means and CoHiRF-KMeans references respectively (mean median of agents across five random feature partitions).

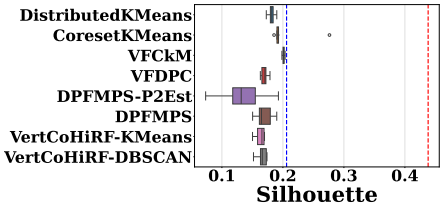
1.2.2 Silhouette



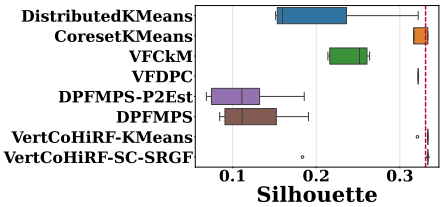
(a) alizadeh-2000-v2



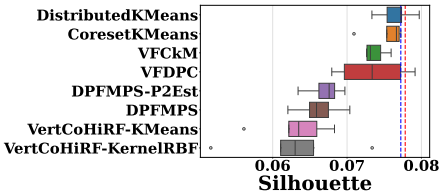
(b) chowdary-2006



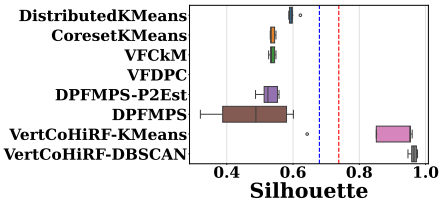
(c) coil-20



(d) garber-2001



(e) nursery

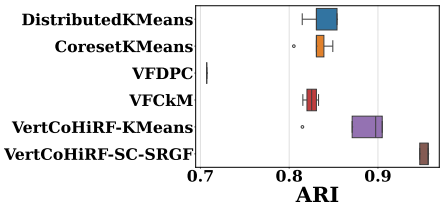


(f) shuttle

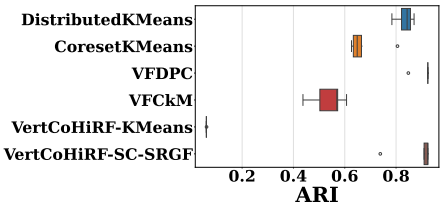
Figure 2: Silhouette boxplots across feature partitions. The blue and red dashed line shows the non-collaborative local k-means and CoHiRF-KMeans references respectively (mean median of agents across five random feature partitions).

2 Robustness in Realistic VFL Scenarios

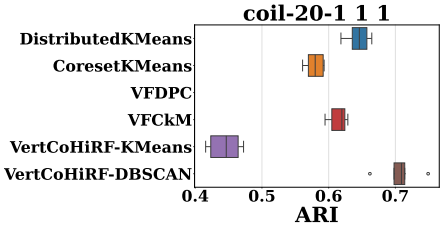
2.1 split weights = [1, 1, 1]



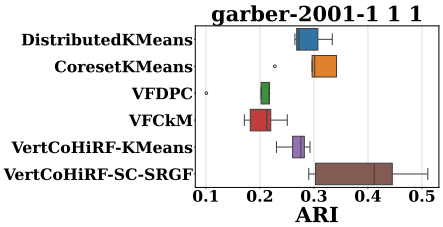
(a) alizadeh-2000-v2



(b) chowdary-2006



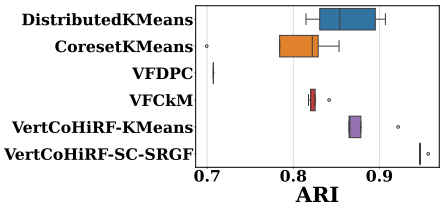
(c) coil-20



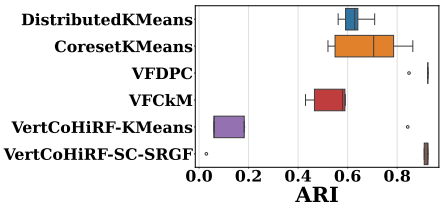
(d) garber-2001

Figure 3: ARI boxplots with a balanced feature partition (importance weights [1, 1, 1].

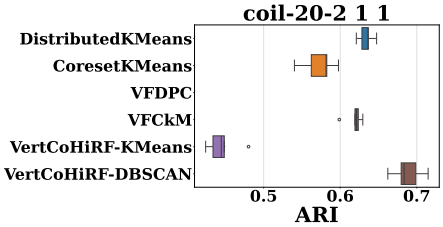
2.2 split weights = [2, 1, 1]



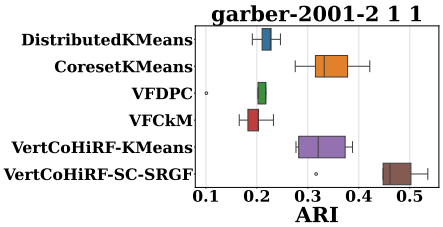
(a) alizadeh-2000-v2



(b) chowdary-2006



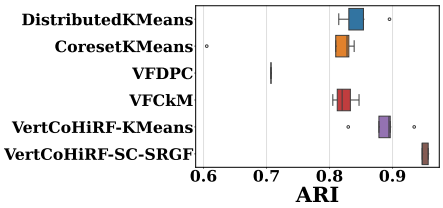
(c) coil-20



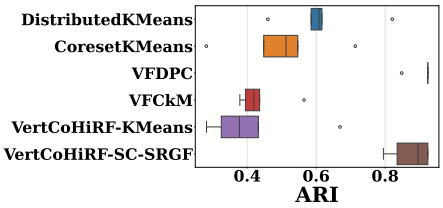
(d) garber-2001

Figure 4: ARI boxplots with a mildly imbalanced feature partition (importance weights [2, 1, 1].

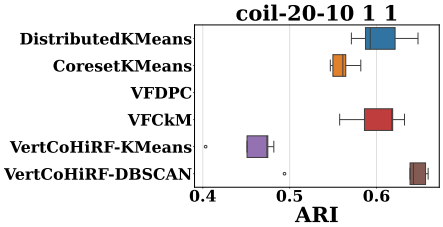
2.3 split weights = [10, 1, 1]



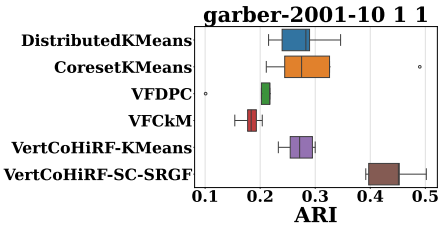
(a) alizadeh-2000-v2



(b) chowdary-2006



(c) coil-20

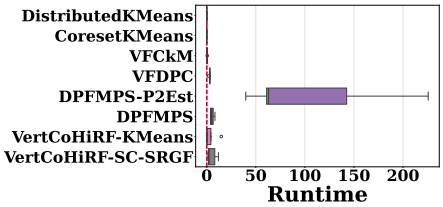


(d) garber-2001

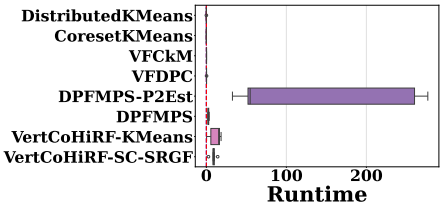
Figure 5: ARI boxplots with a strongly imbalanced feature partition (importance weights [10, 1, 1].

3 Runtime of Methods

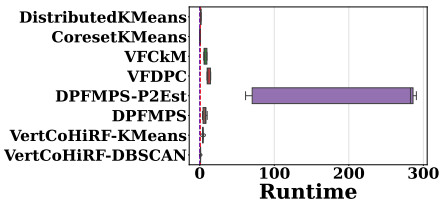
3.1 With DPFMPS/DPFMPS-P2Est



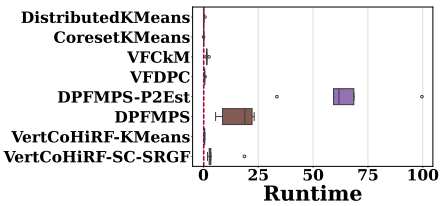
(a) alizadeh-2000-v2



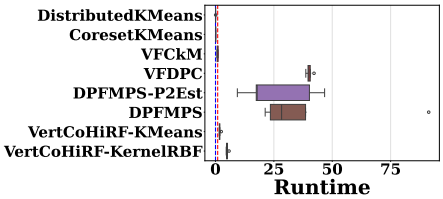
(b) chowdary-2006



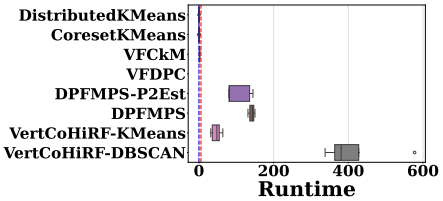
(c) coil-20



(d) garber-2001



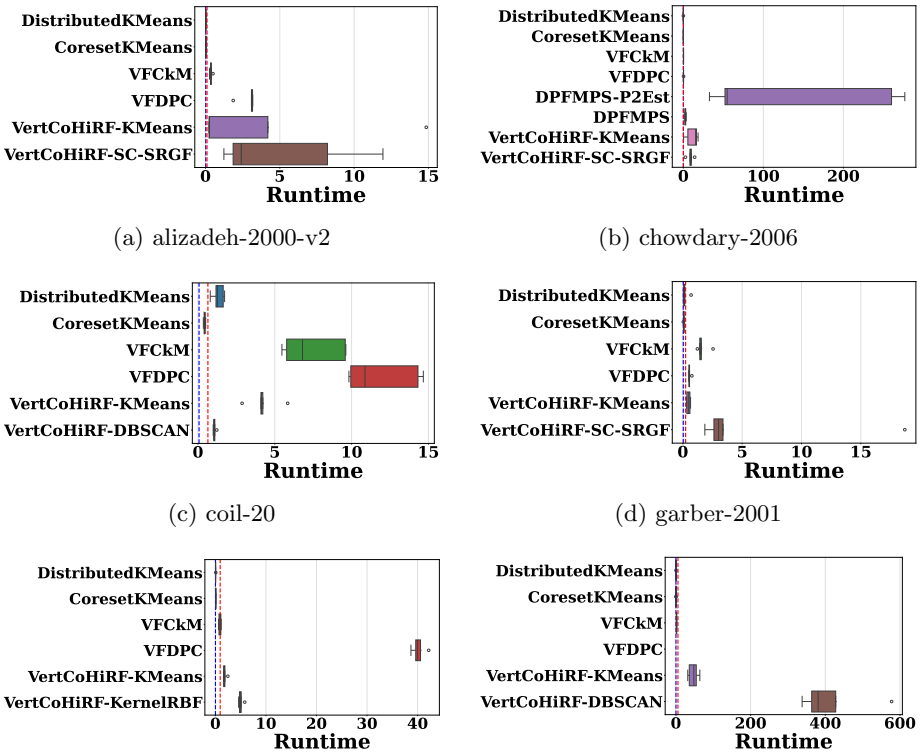
(e) nursery



(f) shuttle

Figure 6: Runtime boxplots across feature partitions. The blue and red dashed line shows the non-collaborative local k-means and CoHiRF-KMeans references respectively (mean median of agents across five random feature partitions).

3.2 Without DPFMPS/DPFMPS-P2Est



(a) alizadeh-2000-v2

(b) chowdary-2006

(c) coil-20

(d) garber-2001

(e) nursery

(f) shuttle

Figure 7: Runtime boxplots across feature partitions without considering the method DPFMPS/DPFMPS-2PEst. The blue and red dashed line shows the non-collaborative local k-means and CoHiRF-KMeans references respectively (mean median of agents across five random feature partitions).

4 Convergence of Active Medoids

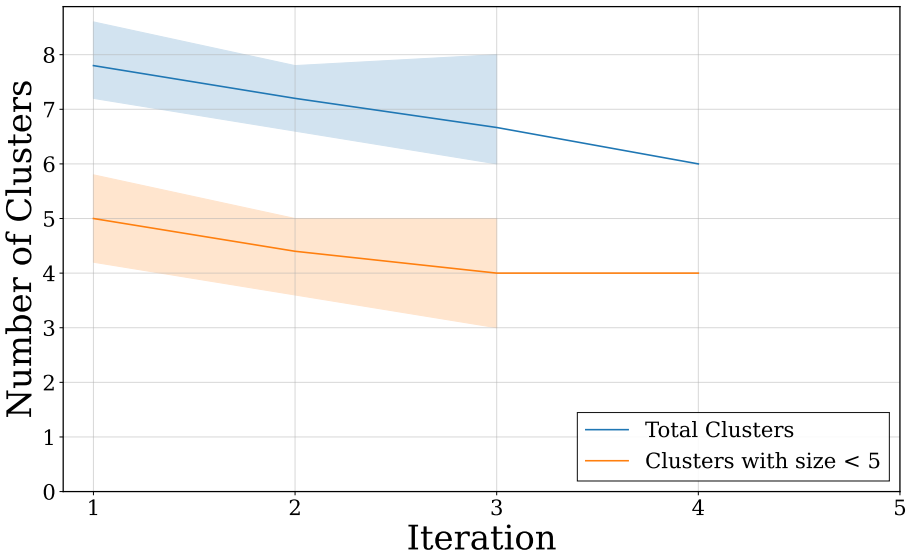


Figure 8: Number of clusters (or active medoids) and number of clusters with less than 5 samples across iterations for VertCoHiRF on the dataset `alizadeh-2000-v2`. We show the mean results across 5 different seeds.